

(WIP) Bridging the Industry-Academia Gap: Assessing the Need for AI Tools in Technical Interview Preparation and Workforce Readiness.

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Abstract—This research-to-practice Work-in-Progress (WIP) paper explores how students engage with Artificial Intelligence (AI) tools, particularly large language models (LLMs), for technical interview preparation and workforce readiness. Despite the increasing availability of AI-driven career platforms, students report limited familiarity with and use of these tools for interview practice. Grounded in AI literacy and Social Cognitive Career Theory (SCCT), this study examines underuse through the lenses of self-efficacy and outcome expectations. Prior research highlights a persistent Industry–Academia Gap (IAG), where graduates often lack career preparedness due to insufficient workforce-aligned training. While LLMs are widely adopted in professional settings, students frequently overlook their potential for career development.

Using a mixed-methods approach, we surveyed students on their use of AI tools for resume building, interview practice, and skill-gap identification. Participants reported which tools they had used, including Big Interview, Interview Warmup, ChatGPT, Bard, and Microsoft Copilot. Our findings revealed minimal adoption of AI tools for career preparation, with higher engagement reported for academic support. These patterns suggest that while students recognize AI's potential, they underutilize it for professional growth.

Our findings also highlight the need to embed AI-supported career readiness into computing curricula and to strengthen collaboration between academia and industry. Future work will focus on designing and evaluating targeted AI interventions to enhance self-efficacy, increase perceived utility, and bridge the Industry–Academia Gap, ultimately improving students' readiness for technical interviews and early career success.

Index Terms—Higher Education, Employability, Computing Skills.

I. INTRODUCTION

As the demands of the computing and engineering workforce evolve rapidly, students are increasingly expected to be proficient not only in technical content but also in career-oriented skills such as interview readiness, resume development, and self-directed learning [1]. However, many academic programs continue to emphasize traditional theory-heavy curricula, often at the expense of hands-on technical training, workforce readiness, and soft skills development [2]. This mismatch between academic preparation and industry expectations constitutes the well-documented Industry-Academic Gap

(IAG), which leaves many recent and prospective graduates struggling with technical interviews, resume building, and identifying relevant skill gaps [3], [4].

AI-driven career tools, particularly Large Language Models (LLMs), offer a promising means of addressing the IAG by providing personalized and adaptive learning experiences [5], [6]. These tools can offer scalable, context-aware guidance that supports students in developing critical career skills such as job search strategies, interview preparation, and professional communication. Traditional resources like resume workshops and mock interviews, while valuable, often lack scalability and accessibility. In contrast, LLMs such as ChatGPT, Bard, and Microsoft Copilot provide real-time, individualized feedback that can supplement traditional career counseling [7]. This potential is explored in Mejias et al.'s “Buddy, Not Crutch” framework, which outlines responsible zero-shot prompting strategies using public LLMs to simulate key stages of technical interviews [8].

Despite the increasing availability of AI-driven career platforms, findings indicate that students have yet to fully adopt these tools for technical interview preparation. Most current usage of AI tools remains academia-focused rather than career-driven [3], [9]. This highlights a critical gap in AI literacy for workforce readiness, where students are not fully leveraging these tools for professional development.

Given these challenges, this study investigates and presents data from a survey that assesses the extent to which students engage with AI tools in preparing for technical interviews and workforce readiness.

II. BACKGROUND

A. AI in Recruitment and Interview Preparation

AI technologies are already transforming the recruitment landscape. On the employer side, companies are already adopting these methodologies to streamline hiring, from AI-based resume screening systems to automated interview analytics [10]. For example, Natural Language Processing (NLP) algorithms and Machine Learning algorithms can now analyze

candidate interviews and evaluate speech content, tone, and even expressions to detect competencies [11].

On the candidate side, AI is being harnessed as a preparation tool [12]. A variety of AI-powered interview practice platforms have emerged in recent years. *Big Interview* is one such platform that combines tutorials with an AI evaluation system where users can receive feedback on metrics such as filler word usage, speaking, eye contact, and tone [13]. Importantly, these AI evaluations focus on delivery aspects; *Big Interview*'s algorithm, for example, can count "um"s and measure speech rate, but it does not yet fully assess the substantive quality of the answer content without preset criteria. Even with such limitations, studies have found value in using the tool [13].

Beyond dedicated interview platforms, generative AI has opened possibilities for self-guided practice. LLMs such as OpenAI's ChatGPT can act as interactive career coaches. For example, a student can prompt ChatGPT with: "Ask me a technical interview question about data structures," answer the question, and then ask, "Give me feedback on my answer." While not perfect, the AI can provide a critique, point out missing considerations, and even role-play follow-up questions. Harvard's Mignone Center for Career Success suggests students use generative AI to brainstorm industry-specific questions, draft answers in the STAR (Situation-Task-Action-Result) format for behavioral questions, and obtain feedback on those answers [11]. In a sense, generative AI can serve as a 24/7 practice partner, allowing students to rehearse answers to a wide range of potential questions and refine their responses through iterative dialogue.

B. Theoretical Frameworks

1) *Social Cognitive Career Theory*: This study draws upon principles of Social Cognitive Career Theory (SCCT), which emphasizes the importance of self-efficacy, outcome expectations, and personal goals in career decision-making and behavior [14]. According to SCCT, students are more likely to engage in career development behaviors if they believe in their abilities (self-efficacy), perceive positive outcomes from those behaviors, and have clear personal goals. In the context of AI tools usage, SCCT helps explain students' motivation to adopt (or avoid) new technologies for professional development. A lack of familiarity or confidence in using AI for career-related tasks may diminish self-related preparation efficacy, while uncertainty about the benefits of such tools may reduce outcome expectations [15]. Additionally, ethical concerns, perceived accessibility, and prior experiences can influence the adoption of such technologies [5].

2) *Technology Acceptance Model*: To investigate the adoption of AI tools in interview preparation, our study is guided by the Technology Acceptance Model (TAM) [16]. TAM provides a framework to understand how users come to accept and use new technologies. At its core, it posits two primary factors: *Perceived Usefulness (PU)* - The degree to which the user believes the technology would enhance their performance - and *Perceived Ease of Use (PEOU)* - the degree to which the technology is free of effort. In the context of AI-

driven interview preparation, *Perceived Usefulness* translates to whether students believe these tools can actually improve their interview outcomes or readiness. This might include helping them solve coding problems faster, communicate more clearly, or reduce their anxiety through mock interviews. *Perceived Ease of Use* translates to the user-friendliness of the AI tool, which might include the intuitiveness of the user interface, the presence of technical glitches, and other factors affecting ease-of-use. The anticipation is that users are more likely to embrace AI interview and career guidance tools if they find them genuinely helpful for improving their skills and if the tools are not complicated to use.

III. METHODOLOGY

A. Participants and Context

Our target population was students at or near the point of entering the technical workforce (upper-division undergraduates and early graduate students). We used a convenience sample drawn from upper-division computing courses.

A total of $N = 46$ completed surveys were analysed. Self-reported academic standing was: freshman ($n = 1$, 2.2%), junior ($n = 23$, 50.0%), senior ($n = 14$, 30.4%), early-entry master's ($n = 3$, 6.5%), first-year doctorate ($n = 1$, 2.2%), and undisclosed ($n = 4$, 8.7%). The majority of respondents majored in computing disciplines (e.g., Computer Science with concentrations in Software Engineering, Information Technology, Bioinformatics, Human-Computer Interaction, AI Robotics and Gaming, Cybersecurity, Data Science, or Systems and Networks), aligning the sample with the study's focus on technical interview preparation. This participant profile provided relevant content for examining AI tool adoption: respondents had an incentive to prepare for technical interviews, and many had firsthand knowledge of the challenges involved.

B. Survey Design

We developed a structured survey questionnaire to investigate the multiple dimensions of AI tool adoption for interview preparation. The survey consisted mostly of closed-ended questions with Likert-scale responses, along with a few open-ended prompts to gather qualitative insights.

The survey covered four key areas: (1) Familiarity with AI tools for resume building, interview preparation, skill-gap identification, and study planning (*Perceived Usefulness and Ease of Use*); (2) AI tool usage, assessing frequency of engagement with ChatGPT, Bard, Gemini, *Big Interview*, Interview Warmup, and Microsoft Copilot (*Perceived Usefulness and Ease of Use*); (3) Attitudes toward AI in technical interview preparation, including its role in problem generation, code optimization, and algorithm selection (*Perceived Usefulness*); and (4) Ethical concerns, evaluating trust, fairness, reliability, and risks of over-reliance on AI for workforce readiness (*User Self-efficacy from the SCCT*).

We designed the survey instrument to capture familiarity, frequency of use, and perceived usefulness of AI tools for interview and academic preparation. Items were written with

neutral wording, avoided double-barrelled phrasing, and were reviewed by domain experts for content validity. We used a 5-point Likert format to balance sensitivity with respondent burden and to retain a neutral midpoint.

IV. RESULTS AND OUTCOMES

A. Awareness and Usage of AI tools for Interview Preparation

TABLE I
FAMILIARITY AND FREQUENCY OF USE OF AI TOOLS FOR INTERVIEW PREPARATION (LIKERT 1–5).

Tools	Familiarity		Usage	
	M	SD	M	SD
ChatGPT	2.69	1.20	2.69	1.22
Bard	1.60	1.09	1.49	0.97
Google Gemini	1.60	1.05	1.64	1.02
Interview WarmUp	1.56	1.10	1.40	0.86
Big Interview	1.47	0.99	1.47	0.97
Microsoft Copilot	1.66	1.08	1.71	1.11

As identified in Table I, mean familiarity and usage for all tools fall below the midpoint of the 1–5 Likert scale, indicating generally low awareness and engagement for interview preparation. ChatGPT is the highest on both familiarity and usage, whereas vendor-specific interview platforms (e.g., Big Interview, Interview WarmUp) and other general-purpose assistants (Bard, Gemini, Copilot) are lower.

The pattern across tools suggests a *parallel familiarity-use* relationship: tools that participants report being more familiar with also tend to be used more frequently, implying that awareness tracks engagement. Standard deviations (Table I) are approximately 1 across measures, indicating substantial between-student variability.

B. Perceived Usefulness and Ease of Use of AI tools

Participants' perceptions of the usefulness of AI tools were largely positive, especially among those who had actual experience with the tools. On a 1 (Strongly Disagree) to 5 (Strongly Agree) scale:

- Towards using AI for interview preparation, participants reported an overall positive attitude, evaluated using the statement *"I have a positive attitude toward using AI in preparing for job interviews."* (M = 3.45, SD = 1.01) on the Likert scale. **86.36%** of participants reported a rating of 3 and above, indicating perceived usefulness and acceptance of the tool in interview preparation.
- Participants acknowledged the prospective effectiveness of AI tools in interview preparation, evaluated using the statement *"AI tools would make me more effective in my interview preparation."* (M = 3.11 and SD = 1.19). **72%** of surveyed participants reported a score of 3 and above. This suggests a generally high perceived usefulness of AI tools for interview preparation. They believe the tools would be effective; however, their familiarity with the tools tells a different story.
- Another notable result supporting the perceived usefulness of AI systems by participants emerges from their

TABLE II
ENGAGEMENT WITH AI TOOLS BY PURPOSE (LIKERT 1–5).

Purpose	M	SD
Academic preparedness	3.20	0.65
Career / interview preparedness	2.04	0.84

responses to the statement *'I use AI chatbots to improve my general language and communication skills'* (M = 3.05, SD = 1.10), suggesting that, overall, students are only somewhat inclined to adopt AI chatbots to improve their communication. However, a closer look reveals that a significant **36%** (15) rated their agreement at 4 or 5 on this element, indicating that these participants consider AI-based software quite helpful in improving language proficiency and communication techniques.

To investigate the extended perceptions of students about the usefulness of the AI tool, we pose questions focused on academic preparation and improvement. For example, participants were asked to personally evaluate statements like *"I use generative AI models to improve my code efficiency and optimize Big O complexity,"* and *"I use generative AI models to aid in algorithm selection and understanding"*. As shown in Table II, students report greater engagement with AI for academic preparedness than for career/interview preparation. The difference is statistically significant (Welch's $t(12) = 3.43$, $p < .01$), reinforcing that AI use is more established for coursework than for job-focused activities.

C. Attitudes, Trust, and Ethics

The attitudinal, trust assessment, and ethical sections of the survey generally revealed favorable views toward adopting AI for interview prep.

- As mentioned above, **86.36%** of the participants reported an above average rating, on the Likert scale, of their agreement with the statement *'I have a positive attitude toward using AI to prepare for job interviews'*. This is a strong endorsement in principle, suggesting that once students are made aware of these tools, they are largely welcomed.
- The statement *"Using AI for interview preparation constitutes cheating"* had a below-average evaluation of **2.3** on a scale of 1 (Strongly Disagree) to 5 (Strongly Agree). This relatively low rating indicates that the majority of participants do not view AI-based interview preparation as dishonest or unethical. Furthermore, it supports the principle of endorsement that raising awareness of these tools and their capabilities in the job preparation space would prospectively increase their usage.
- Evaluating users' ethical stance on AI systems and trust is an extremely important factor to consider in its publicity. We evaluated participants' ethical perceptions of AI systems by presenting them with targeted statements and questions, including *"Which statement best reflects your ethical stance on using AI tools for technical interview preparation?"* and *"To what extent do you trust AI tools*

to make recommendations in your job/internship preparation?”. These items were designed to capture both the normative attitudes (i.e., what participants believe is right or acceptable) and the practical trust they place in AI-driven guidance. 54.2% of surveyed participants responded to the ethical stance by selecting “AI should be used with guidelines to ensure it supports genuine learning and skill development”. However, a substantial subset of them, 26% of the surveyed group, espoused a more permissive view: “AI should be used freely as a learning tool and resource for interview preparation”. This appears to display a perspective that AI should be viewed as a natural extension of available educational resources, further supporting the strong endorsement principle.

V. DISCUSSION AND FUTURE WORK

The results of this study shed light on how AI tools are currently situated in the landscape of technical interview preparation and career development. Broadly, our findings paint a picture of cautious enthusiasm: Students are generally positive about AI’s role in the educational space, yet actual adoption lags behind interest due to awareness.

One of the driving motivations for this study was the persistent gap between academic preparation and industry expectations. With 96% of employers perceiving current preparation as insufficient [17], it is evident that new approaches are needed. The high interest displayed in AI tools’ ability suggests that students themselves recognize that conventional methods alone may not suffice and are open to augmenting their preparation with technology. This can help educational institutions address at least two aspects of the gap:

- **Skill Alignment:** AI-driven practice tools expose students to scenarios and problems that can be closer to what the industry uses. For example, practicing coding challenges or systems design questions via AI better mirrors the format of actual technical interviews (which themselves aim to simulate real work tasks) than, say, studying theoretical questions from textbooks. This form of practice is key to bridging academic knowledge with practical application, similar to how labs or internships function, thus building confidence and relevant skills [18].
- **Volume and Personalization of Practice:** Historically, one limitation in students’ preparation for interviews has been providing enough practice opportunities with feedback [19]. AI tools, however, can deliver unlimited practice for as long as students are willing to engage and give feedback on that. While not as rich as human feedback, it still offers value. This intensity of practice can accelerate skill development and expose them to a wider variety of problems. In essence, AI can help scale up career preparation, an important factor given the need to prepare all students, not just the few who proactively seek help. These ideas align with the ethical prompting patterns introduced by Mejias et al. [8], who propose a structured, student-centered approach to using LLMs as

practice partners, not as crutches, in technical interview preparation.

A. What can educational practitioners and tool designers do to address the barriers identified?

1) *Awareness and Advocacy:* Notably, the adoption of such tools for career development and preparation in higher education is still in the early stages. Many students remain unaware of their existence or have not tried them, and faculty or career advisors may be uncertain about recommending AI tools without understanding their efficacy. A recent literature review highlighted that there is little research specifically focused on integrating AI technologies for career preparation and interview readiness within educational programs [13]. A clear implication is that universities should proactively introduce these tools to students. Additionally, as Burge et al. note, students from underrepresented groups often face institutional barriers that limit exposure to high-impact career preparation experiences [20]. Embedding AI tools in coursework could help mitigate these disparities by providing scalable, just-in-time support.

This study found that many participants discovered tools late or by chance. Career services can also play a big role: for example, advertising the availability of *Big Interview* and *Interview WarmUp* or curating a list of recommended free AI tools (like advising students how to use LinkedIn’s AI feedback feature or ChatGPT effectively).

2) *Improving AI Feedback:* For tool developers, our results highlight areas for improvement. Personalizing feedback (taking into account a user’s progress over time or known weaknesses) could increase perceived usefulness. Another approach is for tools to be transparent about what they can and cannot evaluate. If a tool clearly states, “We evaluate your progress and ability, not content,” students know to seek content feedback elsewhere. This honesty can increase trust in the specific feedback that is given.

The crucial point in our goal was to use technical interview preparation as a strategic entry point for AI in career guidance. We aim to evaluate and identify reasons behind students’ underutilization of AI tools for career guidance and development, design models and tools that highlight how AI can be effectively utilized in the domain of career development, synthesize cognitive adaption, student desire and goals with related AI-Software development, and deliver frameworks on how these AI models and tools can be extended to other areas of career development, such as Resume building and Portfolio feedback, Career Exploration and Advising, Soft Skill development, and Continual Learning and Up-skilling.

VI. LIMITATIONS OF THE STUDY

While our study provides valuable insights, it is important to acknowledge its limitations:

1) *Sample and Generalizability:* Our sample of 46 participants is relatively small and drawn mostly from computing-related fields at one institution. Their experiences and attitudes might not represent all student populations, especially non-technical majors or students in different educational contexts.

2) *Self-Report Bias*: Much of our data is based on self-reported perceptions (usefulness, confidence, etc.). These may be subject to biases such as optimism or the desire to give a “right” answer.

3) *Rapidly Changing Tool Landscape*: The AI tools available are evolving quickly (for instance, updates to ChatGPT or new products launching). Our study captures perceptions at a certain moment, but as tools improve or new tools come online, perceptions may shift. Therefore, some concerns might fade (or new ones emerge) as the AI landscape changes.

VII. CONCLUSION

This research-to-practice investigation set out to explore the need for AI tools in technical interview readiness as a focused case study. The findings provide encouraging evidence that AI has a valuable role to play in career guidance and student professional development. They also suggest that structured AI literacy initiatives and the development of career-focused AI tools may bridge the industry-academia gap (IAG). Students today are open to innovation and, as our study shows, they largely welcome AI-driven support in their journey to become workforce-ready. When properly introduced and guided, AI tools can empower students to take charge of their preparation through intensive practice, immediate feedback, and personalized skill-building. AI tools should be viewed not as a silver bullet but as a potent supplement to traditional training.

For educators and institutions deliberating on whether to embrace AI in their career support offerings, the time is ripe to innovate. Ignoring these tools would mean leaving a beneficial resource untapped and potentially letting the industry-academia gap widen further.

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